What matters for central bankers and what matters for the financial market?

Preliminary results of a Latent Dirichlet Allocation analysis of ECB speeches and data analysis of European countries.

What matters for central bankers and what matters for the financial market?

Why is this important?

- Understanding the relationship between **market signals and central bank priorities**
- helps explain how policy and market perceptions interact.
- If financial markets and policymakers focus on **different risks**,
- this may lead to **mispricing, policy uncertainty, or weaker policy transmission**.
- Identifying these **differences improves our understanding of**
- **monetary policy credibility and financial market reactions.**

Strategy

- **Financial market**
- **Sovereign spreads** signal how markets evaluate macroeconomic fundamentals such as
- GDP growth, competitiveness (REER), fiscal sustainability (Debt/GDP) , and international vouneralability (Currrent account)
- **Central bankers**
- **Speeches** serve as a proxy for the issues policymakers consider most relevant.

Observations and Average GDP Growth

- Our analysis focused on 11 countries in the EU: Portugal, Lithuania, Italy, France, Finland, Spain, Germany, Austria, Ireland, Greece, Belgium, and the Netherlands.
- Most countries (7) had 800 observations in the dataset, whereas the others had 784 (Ireland and Greece), 776 (Belgium), and 774 (the Netherlands).
- Our analysis also shows the average GDP. Almost all countries' GDP growth is below 1, except Ireland's, whose GDP growth is 1.24.

Unnamed: 0	country	observations
0	PT	800
1	LU	800
2	IT	800
3	FR	800
4	FI	800
5	ES	800
6	DE	800
7	AT	800
8	IE	784
9	GR	784
10	BE	776
11	NL	774

Unnamed: 0	country	avg_gdp_growth
0	IE	1.24239
1	LU	0.553625
2	ES	0.408339
3	BE	0.385948
4	NL	0.375239
5	AT	0.331959
6	FR	0.309947
7	FI	0.277044
8	DE	0.248734
9	PT	0.239222
10	GR	0.168351
11	IT	0.102007

Average Spread in the Data

- Our analysis focused on the average spread. The country with the highest average spread was Greece (around 3.92), whereas the country with the lowest spread was Germany (0). Both the highest and lowest spread countries are significantly different from the other countries, whose spread is around 0.5-1.7.
- Our analysis also shows the countries whose average spread is above the overall average spread in the dataset. Namely, these countries are Spain, Greece, Ireland, Italy, and Portugal.

Unnamed: 0	country	avg_spread
0	GR	3.924514
1	PT	1.760314
2	IT	1.336504
3	IE	1.070474
4	ES	1.026203
5	BE	0.488914
6	FR	0.363014
7	AT	0.348723
8	FI	0.275042
9	NL	0.204464
10	LU	0.100724
11	DE	0.0

Unnamed: 0	country	avg_spread
0	ES	1.026203
1	GR	3.924514
2	IE	1.070474
3	IT	1.336504
4	PT	1.760314

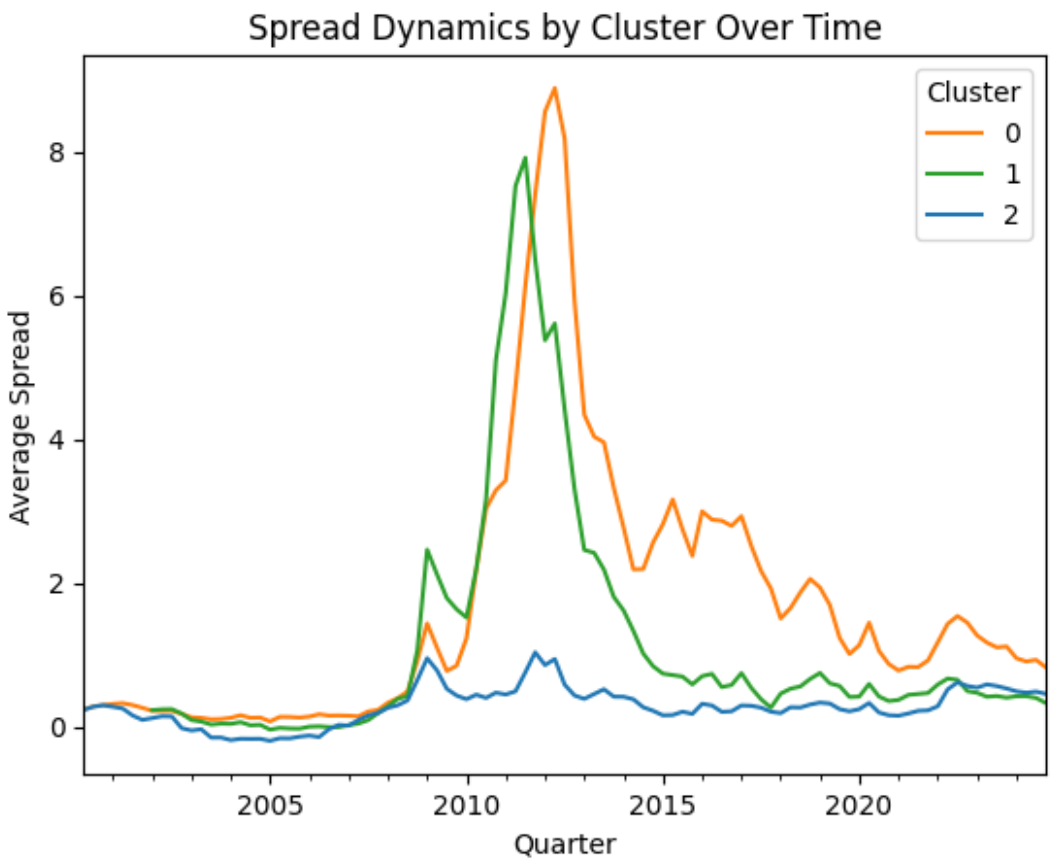
Debt, GDP, and Spread

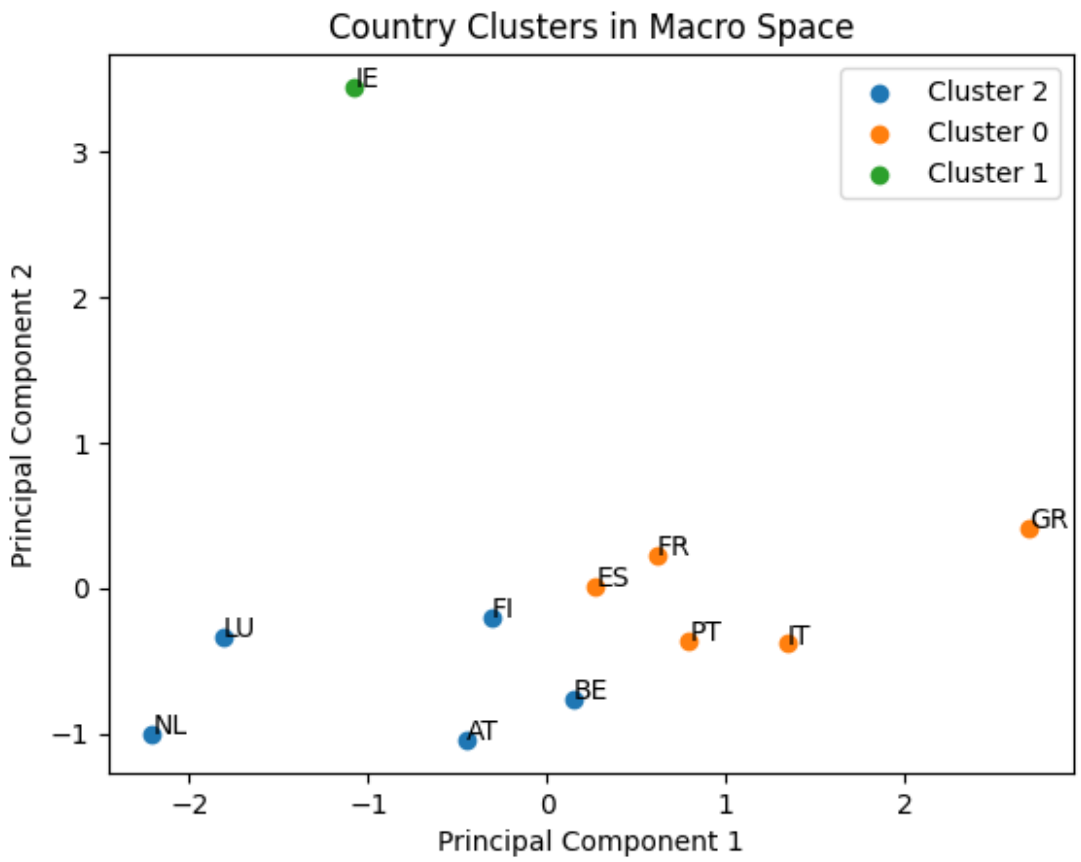
- The country with the lowest debt-to-GDP ratio was Austria in 2021 Q1 (87). On the other hand, the country with the highest debt-to-GDP ratio was Portugal in 2011 Q1 (101.1).
- The country with the highest spread was Austria in 2012 Q1 (1.2), whereas the country with the lowest spread was Portugal (-0.002) in 2005 Q1.

Unnamed: 0	country	date	debt_gdp	debt_rank
0	AT	2021Q1	87.0	1
1	AT	2015Q3	86.8	2
2	AT	2015Q2	86.7	3
3	AT	2016Q1	86.2	4
4	AT	2021Q2	85.9	5
...	...	...	...	...
1195	PT	2022Q4	111.2	96
1196	PT	2023Q1	110.8	97
1197	PT	2023Q2	108.2	98
1198	PT	2023Q3	105.4	99
1199	PT	2011Q1	101.1	100

Unnamed: 0	country	date	spread	spread_rank
0	AT	2012Q1	1.2119772005772005	1
1	AT	2011Q4	1.19130303030303	2
2	AT	2012Q2	1.1147983671299466	3
3	AT	2009Q1	1.0503340548340567	4
4	AT	2009Q2	0.8301212121212127	5
...	...	...	...	...
1195	PT	2004Q1	0.0851897860593506	96
1196	PT	2003Q1	0.0848412698412701	97
1197	PT	2004Q4	0.0784199134199128	98
1198	PT	2005Q2	0.0457215007215001	99
1199	PT	2005Q1	-0.0016507936507932	100

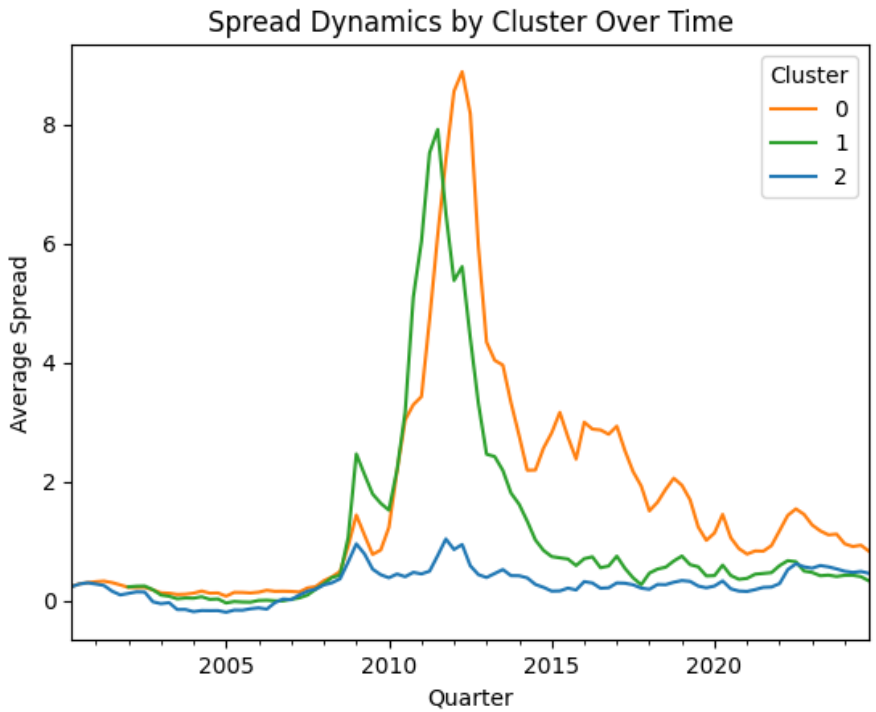
First results I

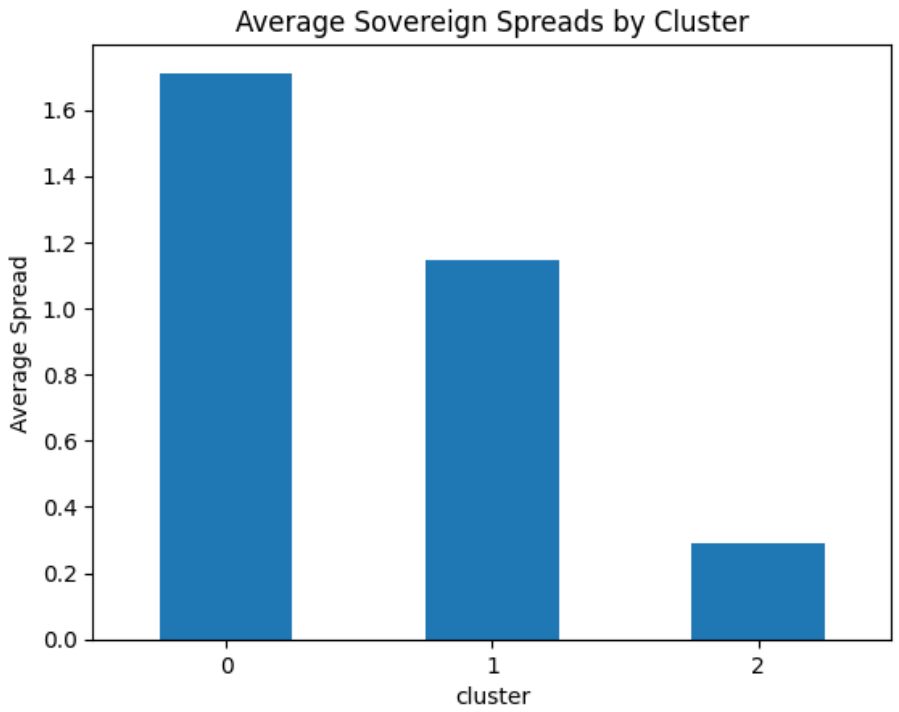




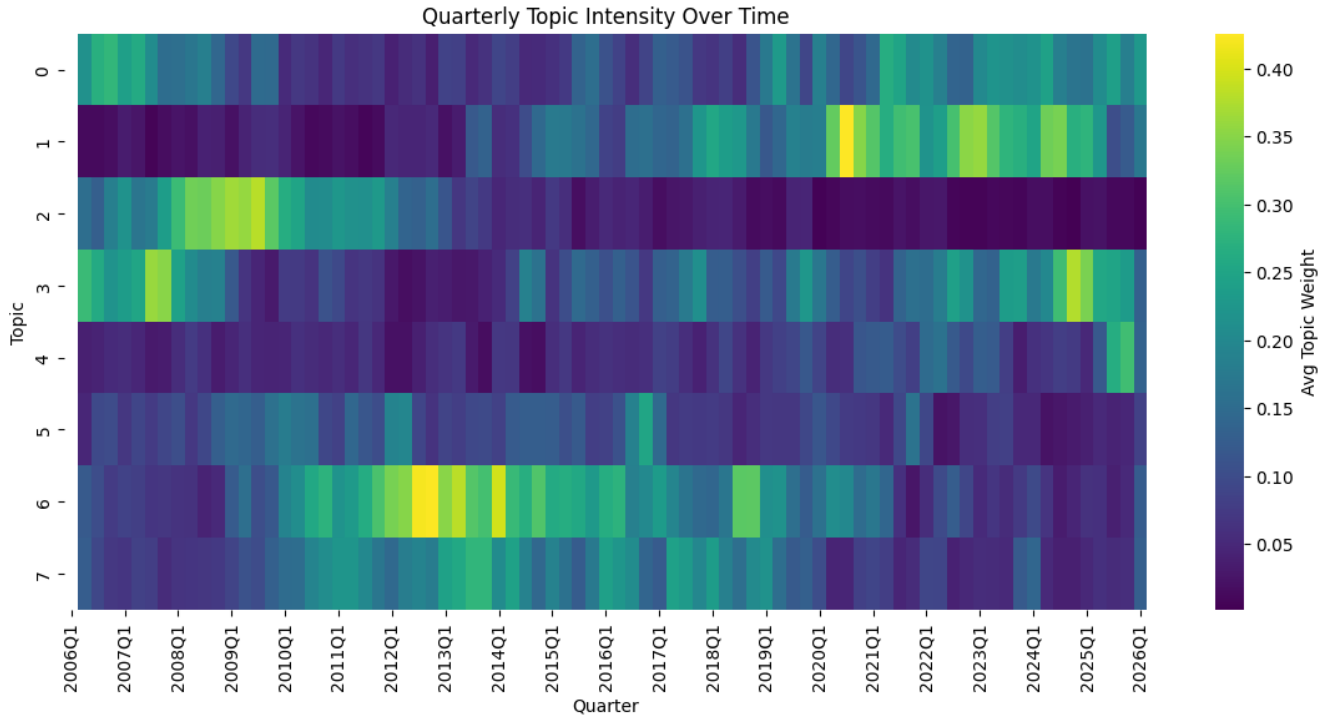
First results II

Cluster **0** exhibits the highest average sovereign spreads and the strongest spike during the sovereign debt crisis, indicating that these countries are perceived by markets as relatively riskier borrowers. In contrast, **Cluster 2** maintains consistently low spreads over time, while **Cluster 1** occupies an intermediate position, suggesting moderate risk levels between the high-spread and low-spread country groups.





Central Bankers Topics

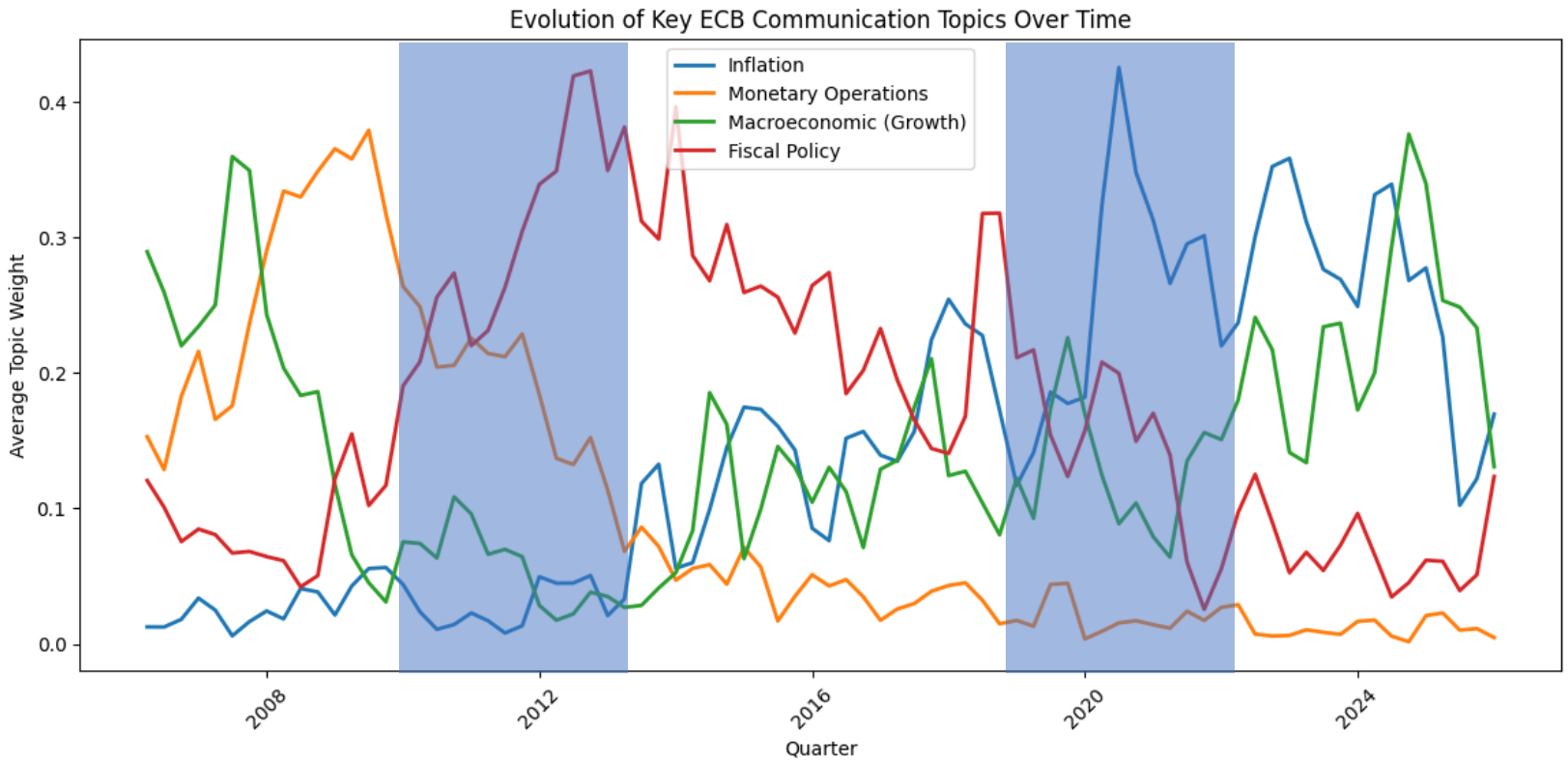


Topic 0	Global Integration
Topic 1	Inflation
Topic 2	Monetary opperations
Topic 3	Macroeconomics
Topic 4	Digital Euro
Topic 5	Financial Policy
Topic 6	EMU Governance
Topic 7	Banking Supervision

Central Bankers Topics

Central bank communication clearly adapts to changing macroeconomic conditions and major economic shocks. The evolution of topic weights shows that the focus of speeches shifts substantially over time, reflecting the challenges faced by policymakers. In particular, the dynamics appear to be **largely driven by monetary and fiscal policy topics**, which dominate the communication agenda during periods of economic stress.

Overall, the results suggest that ECB communication is **highly responsive to the economic environment**, with policymakers adjusting the thematic emphasis of their speeches to address the most pressing economic risks and policy challenges



Period I
Aftermath of the Great Financial Crisis and the onset of the European sovereign debt crisis (2012).

Period II
COVID-19 (2020) and the energy price shock (2022) , leading to elevated inflation in Europe.

Next steps

1. **Vectorize the results:** Construct time series of topic intensities from central bank speeches.

2. **Estimate dynamic models** to study the relationship between speech topics and sovereign spreads, e.g.:

   VAR / SVAR models- Local projections (Jordà impulse responses) Panel VAR across countries

3. **Identify similarities and differences** in how financial markets and central bankers respond to economic shocks.

4. **Contribute to the ongoing debate** on whether a **systematic gap exists between central bank priorities and financial market perceptions**.